

E. Jeffrey Samuels (192524412)

SET 14 – SURVEY RESULTS ANALYSIS

Aim

To analyze survey results using stacked bar chart, grouped bar chart, and table in R.

Algorithm

1. Create the survey results dataset.
2. Convert the questions into long format.
3. Create a stacked bar chart for Question 1.
4. Create a grouped bar chart for all questions.
5. Display the survey response data in a table.

R Program

```
library(ggplot2)
```

```
library(tidyr)
```

```
survey <- data.frame(
```

```
  ID=c(1,2,3),
```

```
  Question1=c("A","B","C"),
```

```
  Question2=c("B","A","A"),
```

```
  Question3=c("C","D","B")
```

```
)
```

```
long <- pivot_longer(survey, -ID,
```

```
  names_to="Question", values_to="Answer")
```

```
ggplot(long[long$Question=="Question1",],
```

```
  aes(Answer, fill=Answer)) +
```

```
geom_bar() +
```

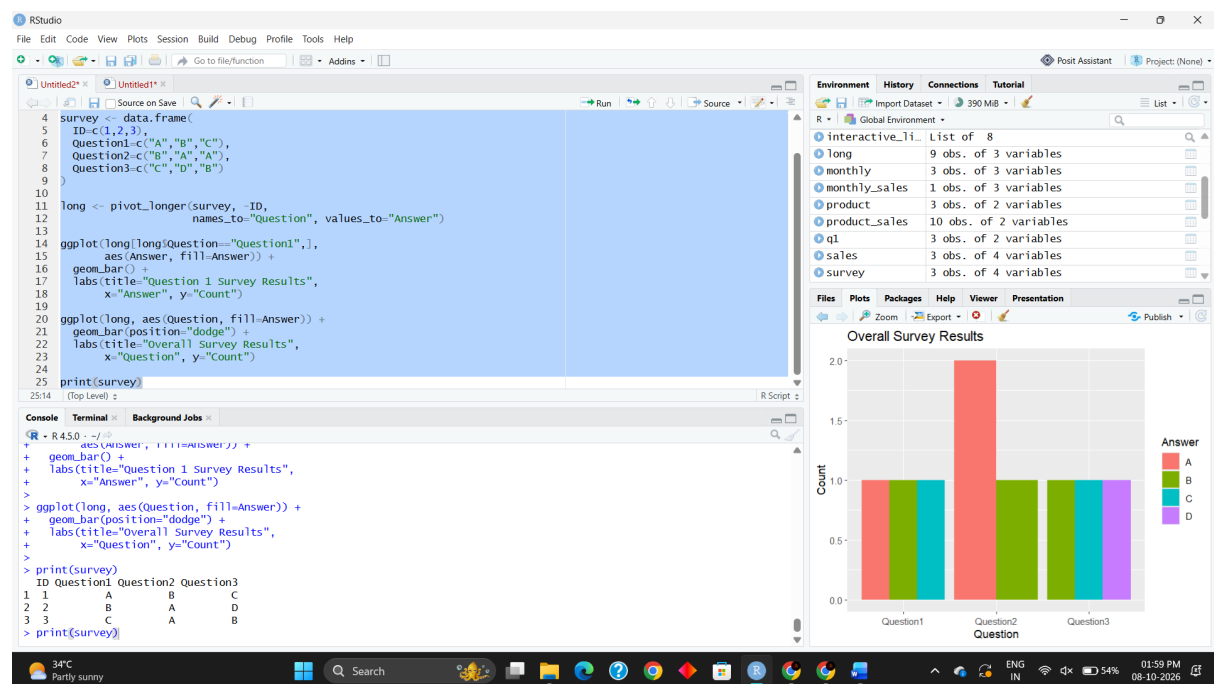
```
labs(title="Question 1 Survey Results",
```

```
  x="Answer", y="Count")
```

```
ggplot(long, aes(Question, fill=Answer)) +
  geom_bar(position="dodge") +
  labs(title="Overall Survey Results",
        x="Question", y="Count")
```

```
print(survey)
```

OUTPUT:



Result

The stacked/grouped charts successfully display the distribution of survey responses for the questions. The table presents the survey response data for each respondent.